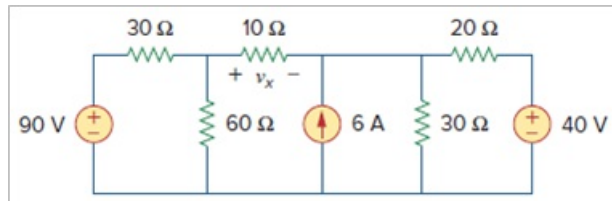


Problem

Use superposition to obtain v_x in the circuit of Fig. 4.85. Check your result using *PSpice* or *MultiSim*.

Figure. 4.85:



Step-by-step solution

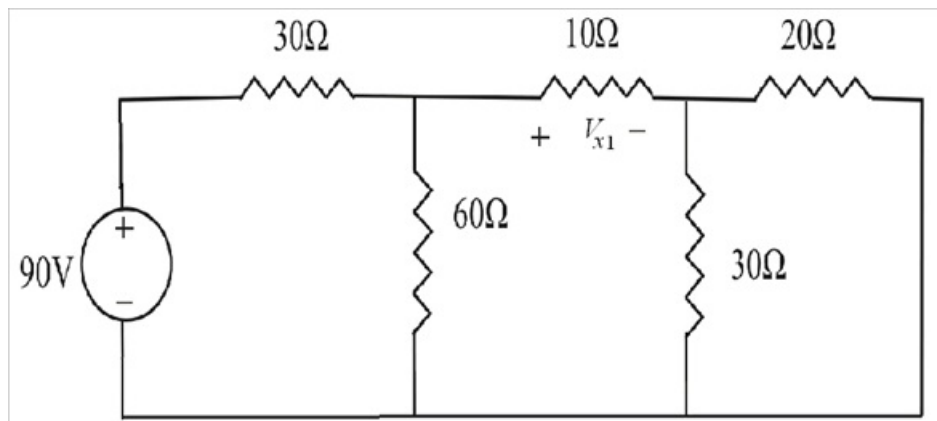
Step 1 of 10

Let $V_x = V_{x1} + V_{x2} + V_{x3}$

Where V_{x1} , V_{x2} and V_{x3} are voltages due to the 90V, 6A and 40V sources alone.

Step 2 of 10

For V_{x1} , consider the circuit below with 90V source alone



Step 3 of 10

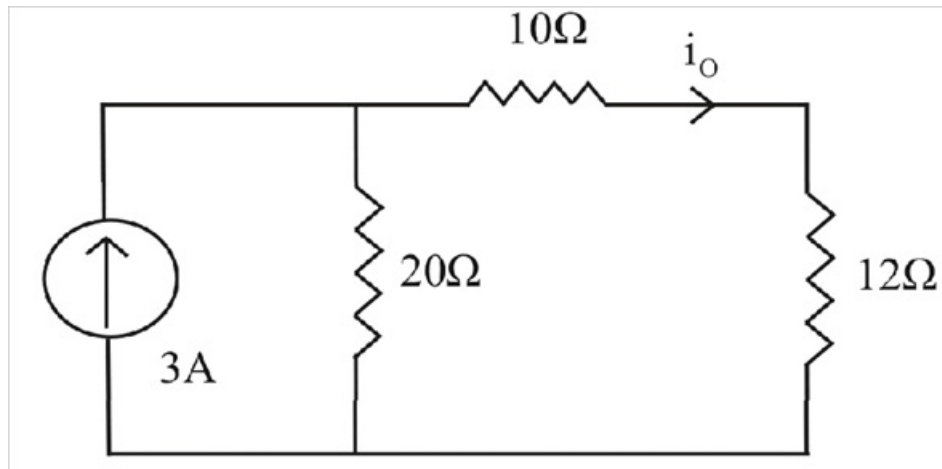
$$20 \parallel 30 = \frac{20 \times 30}{20 + 30}$$

$$= 12\Omega$$

Now by using voltage to current source transformation we can modify the circuit as below:

$$\text{Here } 30 \parallel 60 = \frac{30 \times 60}{30 + 60}$$

$$= 20\Omega$$



Step 4 of 10

From this by using current division rule

$$i_o = 3 \times \left[\frac{20}{20 + 10 + 12} \right]$$

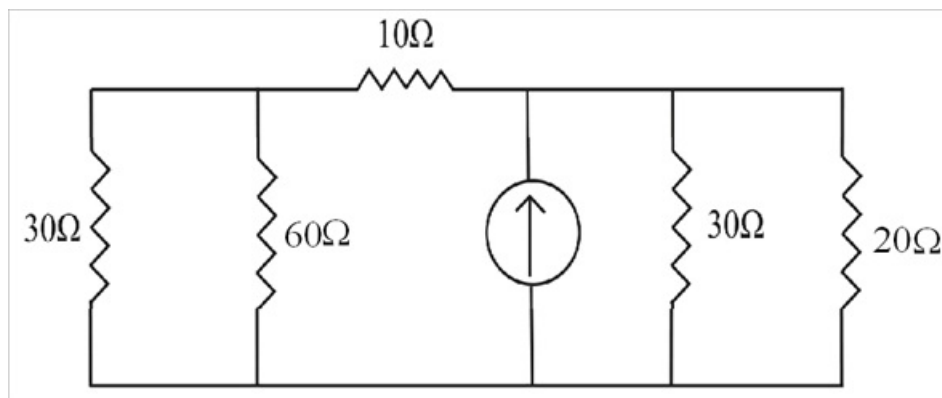
$$i_o = 1.4286\text{A}$$

Therefore voltage across 10Ω resistor $V_{x1} = 1.4286 \times 10$

$$V_{x1} = 14.286\text{V}$$

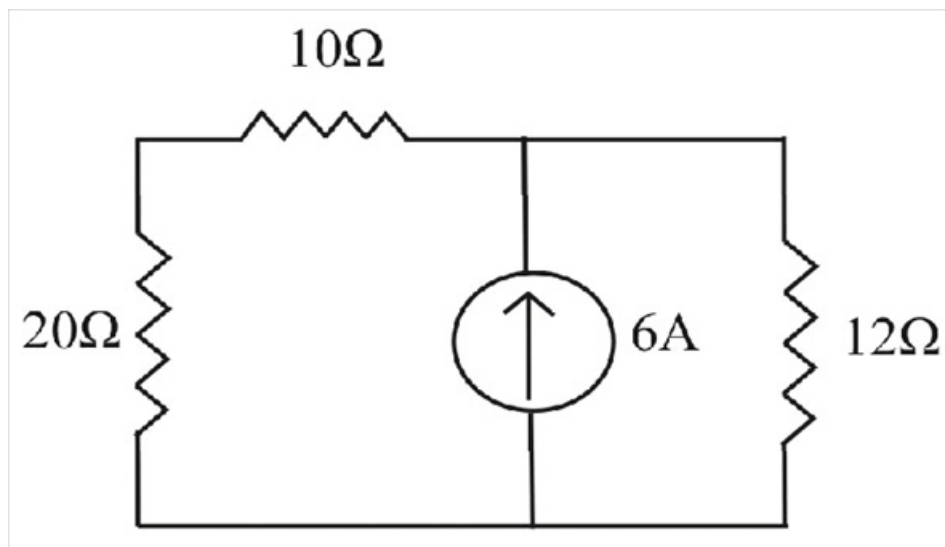
Step 5 of 10

For V_{x2} , consider the circuit below with 6A source alone.



Step 6 of 10

This can be simplified as below:



Step 7 of 10

From by using current division rule current flowing through 10Ω resistor $= 6 \times \left[\frac{12}{12 + 20 + 10} \right]$

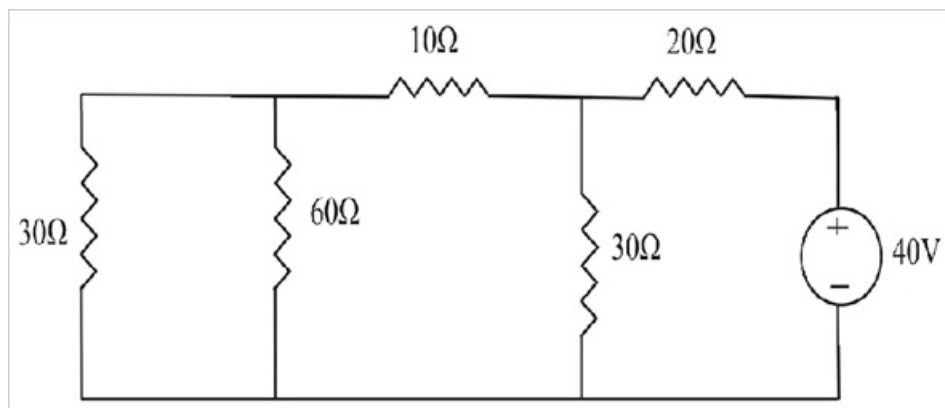
$$= \frac{72}{42} \text{ A}$$

Now voltage across 10Ω resistor $V_{x2} = -\frac{72}{42} \times 10$

$$V_{x2} = -17.143 \text{ V}$$

Step 8 of 10

For V_{x3} , consider the circuit below with 40V source alone.



Step 9 of 10

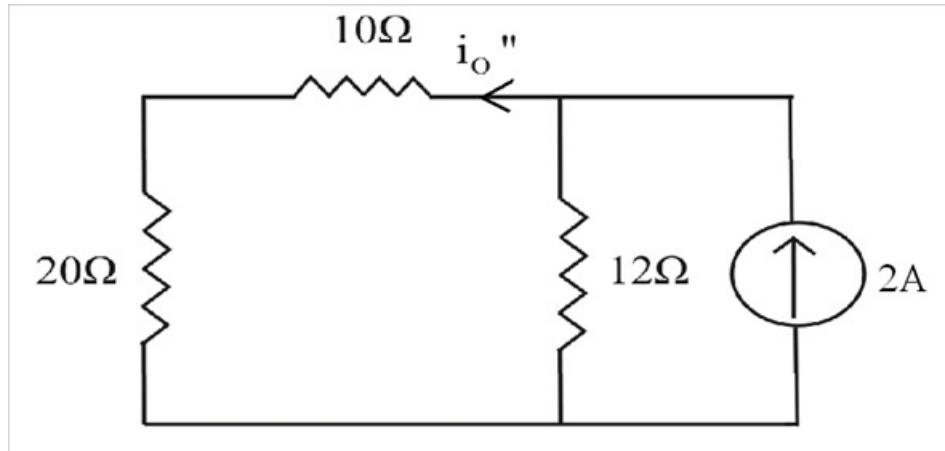
From this $30 \parallel 60 = \frac{30 \times 60}{30 + 60}$

$= 20\Omega$

By using voltage to current source transformation we can modify the circuit as below:

Hear $20 \parallel 30 = \frac{20 \times 30}{20 + 30}$

$= 12\Omega$



Step 10 of 10

By using current division rule

Current flowing through 10Ω resistor $= 2 \times \left[\frac{12}{12 + 20 + 10} \right]$

$= \frac{24}{42} A$

Voltage across 10Ω resistor $V_{x3} = -\frac{24}{42} \times 10$

$V_{x3} = -5.7143V$

Total voltage due to three sources

$V_x = 14.286 - 17.143 - 5.7143$

$V_x = \boxed{-8.5713V}$